Optimal Transport Under Land Cover Information Constraints: Fusing Heterogeneous SAR Imagery and Social Media Data

Zhenjie Liu, *Member, IEEE*, Jun Li, *Fellow, IEEE*, Lizhe Wang, *Fellow, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— The integration of remote sensing and citizen science offers an unprecedented opportunity for observing the Earth and human activities. Recently, heterogeneous data fusion models have been proposed to align representations and geolocations of remote sensing imagery and social media data, such as optimal transport (OT) and geographic OT (GOT). However, these models generally ignore the differences in remote sensing features of the same geographical phenomenon for different land cover types, which not only affects the fusion accuracy but also leads to the loss of fusion information. In this study, we develop a general model for heterogeneous SAR imagery and social media data fusion based on OT and land cover information, namely, land cover information-constraint OT (LCIOT). Taking the 2017 Houston flood event as a case study, the experimental findings demonstrate that the proposed LCIOT can align 99% of geotagged Twitter data to flooded areas with an average transport distance of 710 m, outperforming the state-of-the-art models, i.e., OT and GOT. By combining the supervised information obtained by LCIOT with the SAR imagery, the overall accuracy (OA) and Kappa of urban flood mapping in the three study areas ranges from 0.77 to 0.82 and 0.51 to 0.61, respectively. Overall, the proposed LCIOT provides a new perspective to solve the issues of discrepancies in data distribution and geolocation uncertainty in the context of heterogeneous data fusion.**

***Index Terms*— Data fusion, flood mapping, land cover, remote sensing, social media.**

I. Introduction

In THE 21st century, remote sensing and citizen science are considered promising solutions for describing the complexity of the Earth system, especially in the applications of smart cities [1], emergency response [2], environmental monitoring [3], etc. As a powerful technology for Earth observation, remote sensing involves the use of systems and algorithms to record information about the Earth surface from a remote location [4]. In the context of citizen science, volunteer geographic information (VGI) derived from social networks can provide real-time and rich information about the local environment [5]. The fusion of global-view remote sensing data and local VGI provides a new perspective for achieving more reliable and accurate information and generating improved insight. However, since the data collected from remote sensing and citizen science are of different modalities, it is a typically challenging task for heterogeneous data fusion.

With an ever-growing number of satellites in orbit, massive remote sensing data have been produced with different spectral, spatial, and temporal resolutions [6]. Optical sensors and synthetic aperture radar (SAR) are the two primary sources of remote sensing imagery. Optical imagery can provide rich spectral information and enhance the contrast of land cover types through band reflectivity differences, but its availability is limited by cloud cover [7]. In comparison, the all-weather and all-day information acquisition capabilities make SAR usable even in extreme weather conditions [8]. Despite the wide availability of massive remote sensing datasets from numerous satellite systems, it may be difficult to gather the required data within a specific spatiotemporal range due to the limitations of the satellite revisit cycle, atmospheric opacity, and other obstructions.

With the development of the internet and Web 2.0, social media provides one of the most important data sources for VGI in a real-time manner, which has been proven to be a meaningful addition to remote sensing data [9]. Social media data are easily created by the social networks (e.g., Twitter, Facebook, YouTube) of smart mobile devices in the form of text, photographs, and videos. In particular, geotagged data with latitude and longitude coordinates are the most popular source to obtain in situ information [10]. The immediacy and low cost of social media data provide new opportunities for observing human activities, but their noisy nature and quality uncertainty pose additional challenges.

The fusion of remote sensing imagery and social media data needs to solve the issues of discrepancies in data distribution and geolocation uncertainty [11], [12]. On one hand, these data are heterogeneous due to differences in the data generation process, causing discrepancies in the probability density

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functions (pdfs) [13]. To solve this issue, it is essential to establish a mapping based on the physical or statistical relationship between the data [14]. On the other hand, most existing studies on the fusion of remote sensing imagery and social media data ignore the issue of location uncertainty, which refers to intentional or unintentional inconsistency between geographic entities and geographic phenomena [15], [16]. For instance, in a flood event, social media users often post messages from nonflooded locations to indicate flooding phenomena, resulting in a mismatch between social media flooding data and flooded locations in remote sensing imagery [17].

Recently, some studies have put forward heterogeneous data fusion models based on domain adaptation to align representations and geolocations of remote sensing imagery and social media data [11], [18]. Domain adaptation is a technique related to machine learning and transfer learning that aims at learning a model from a source distribution to perform well on a target distribution, such as optimal transport (OT) [19], [20]. The research in Wang et al. [11] proposed a multimodal fusion framework for Landsat imagery and Twitter data to address inherent sensor noise and develop density estimation maps of flood disasters based on OT. On this basis, Liu et al. [21] combined the normalized difference vegetation index (NDVI) and the normalized difference water index (NDWI) derived from remote sensing imagery and developed a geographic OT (GOT) model to fuse heterogeneous data. These models rely on the remote sensing features that are able to characterize specific geographic phenomena. There might be significant differences in the remote sensing features of the same geographical phenomenon for different land cover types, which are generally ignored when constructing the association between geographic entities and geographic phenomena. In this case, the fusion accuracy will be affected, and the loss of fusion information will occur. Moreover, the existing models are mainly designed to fuse optical imagery and social media data, but the fusion of SAR imagery and social media data needs to be further explored.

Based on the aforementioned research and deficiencies, we develop a general model for heterogeneous data fusion based on OT and land cover information, called LCIOT. In addition to solving the issues of discrepancies in data distribution and geolocation uncertainty, LCIOT aims at simultaneously aligning social media data to geographic phenomena with different remote sensing features caused by various land cover types. The main innovative contributions of our work can be summarized as follows.

1) A general data fusion model is developed for heterogeneous SAR imagery and social media data.
2) Variations in backscatter and coherence for different land cover types are introduced as constraints in LCIOT.
3) The proposed LCIOT can generate comprehensive and reliable samples for supervised classification tasks.

The remainder of this article is organized as follows. Section II introduces some related works. Section III describes the newly proposed LCIOT model in detail. Section IV describes the datasets used in our study. Section V presents the experimental results. Section VI discusses the results and provides hints at plausible future research lines. Section VII concludes this article with some remarks.

II. Related Works

In this section, we review the existing literature on domain adaptation and OT, and the fusion of remote sensing imagery and social media data.

A. Domain Adaptation and OT

Current research on domain adaptation can be mainly divided into three categories according to the methods used: sample-based methods [22], feature-based methods [23], and inference-based methods [24]. As one of the most popular feature-based domain adaptation methods, OT describes the optimal way to transport data between source and target distributions by considering the transport cost from one space to another, which has been widely adopted in the fields of image processing, computer graphics, and machine learning [25], [26]. Recently, OT has been adopted in multisource data fusion. For instance, Courty et al. demonstrated the practical application of OT in three types of remote sensing data fusion problems: domain adaptation, time-series averaging, and change detection in LIDAR data [13]. Wu et al. [27] proposed a point–surface data optimal fusion method to fuse geological survey data and geological remote sensing interpretation (GRSI) products based on OT. Mifdal et al. [28] presented an application of OT in the fusion of spectral information of the hypespectral image and the spatial information of the multispectral image.

B. Fusion of Remote Sensing Imagery and Social Media Data

Remote sensing imagery and social media data are highly complementary in Earth observation missions, and a large number of studies have developed data fusion methods suitable for different scenarios [29], [30], [31]. The methods based on semantic information extract information from social media texts through natural language processing (NLP), machine learning, and text mining to correlate or enhance remote sensing observations [32], [33]. For example, Ahmad et al. [34] developed a system called "JORD" that automatically collected social media data related to natural disasters by filtering time and text and linked social media data with remote sensing imagery through NLP-based text analysis. Huang et al. [35] integrated social media, remote sensing, and Wikipedia data through data mining and text mining to track the spatiotemporal patterns of different natural disasters.

The methods based on location information use density estimation or spatial interpolation to process social media data and then perform spatial overlay analysis with remote sensing imagery [36], [37]. For example, Schnebele and Cervone [38] used kernel density smoothing to generate 2-D layers of volunteered geographical data, which were then weighted and overlaid with remote sensing imagery to calculate flood inundation probability. Cervone et al. [39] synthesized the flood areas extracted from Landsat-8 OLI imagery, geotagged tweets, and geolocated photographs to assess the impact of hurricanes on urban transportation infrastructure.

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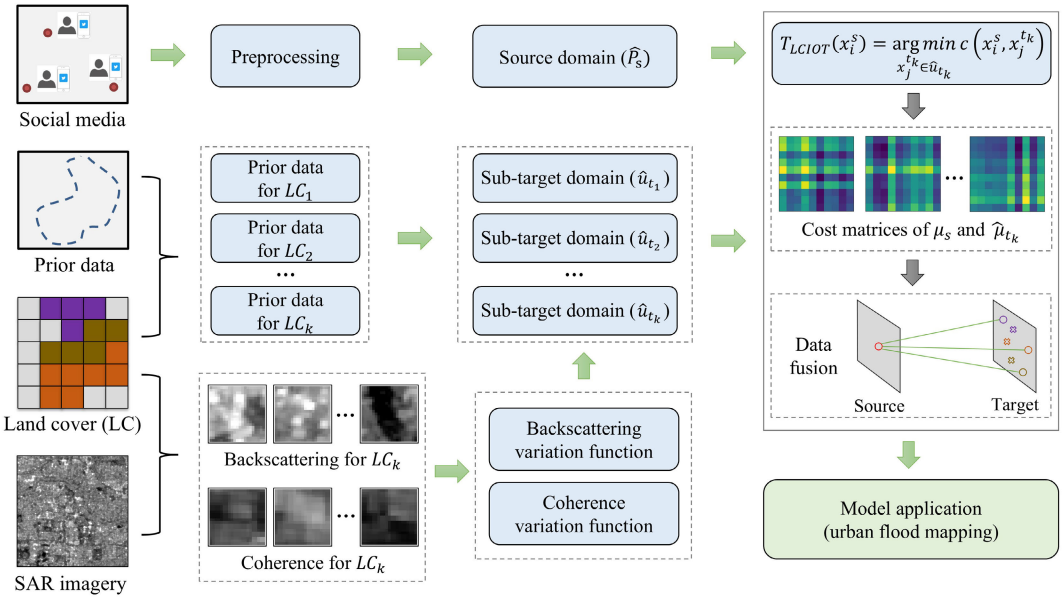


Fig. 1. Flowchart of the LCIOT model for fusing heterogeneous SAR imagery and social media data.

III. Methodology

This section introduces our newly developed LCIOT model for the fusion of heterogeneous SAR imagery and social media data. Fig. 1 shows a schematic overview of the LCIOT. The input data for the LCIOT include social media data, SAR imagery, prior data, and land cover data. We first split the target domain of prior data according to land cover data as subtarget domains. The elements in the subtarget domains are then filtered by the backscattering variation function (BVF) and coherence variation function (CVF). The geographical local correlation mode is adopted to transport the source elements from social media data to each subtarget domain. After data fusion, the application of LCIOT to supervised classification tasks is considered, such as urban flood mapping.

A. Land Cover Information-Constraint OT

Similar to OT [11] and GOT [21], the proposed LCIOT is a heterogeneous data fusion model based on domain adaptation. Specifically, the source and target domains are, respectively, denoted as $D_s = \{x_1^s, \ldots, x_{n_s}^s\}$ (i.e., social media data) and $D_t = \{x_1^t, \ldots, x_{n_t}^t\}$ (i.e., prior data), with n_s and n_t being the number of elements in D_s and D_t, respectively.

Assume there are K land cover types derived from land cover data, D_t can be split into K subtarget domains through spatial overlay analysis, and the k subtarget domain is expressed as $D_{t_k} = \{x_1^{t_k}, \ldots, x_{n_{t_k}}^{t_k}\}$, with n_{t_k} being the number of elements in D_{t_k}.

Let Σ be the σ-algebra on the information fusion space $D = \{x_1, \ldots, x_n\}$, and $\mathcal{P}(D)$ denotes the space of probability measures over Σ. The data distribution of the source domain ($P_s \in \mathcal{P}(D)$) and the data distribution of the subtarget domains ($P_{t_k} \in \mathcal{P}(D)$) can be expressed as follows:

$$P_s = \sum_{x_i \in D} a_i^s \delta_{x_i}, \quad P_{t_k} = \sum_{x_i \in D} a_i^{t_k} \delta_{x_i} \tag{1}$$

where δ_{x_i} is the Dirac unit mass at location x_i, and a_i^s and $a_i^{t_k}$ are the coefficients in the probability simplex, i.e., $\sum_{i=1}^n a_i^s = 1$ and $\sum_{i=1}^n a_i^{t_k} = 1$.

As social media data can be accessible through discrete points using recorded latitude and longitude information, and the prior data processed by land cover data can be associated with grid cell centers, P_s and P_{t_k} are estimated through discrete samples in the corresponding distribution, i.e., $\widehat{P}_s$ and $\widehat{P}_{t_k}$

$$\mu_s = \sum_{i=1}^{n_s} \hat{a}_i^s \delta_{x_i^s}, \quad \mu_{t_k} = \sum_{i=1}^{n_{t_k}} \hat{a}_i^{t_k} \delta_{x_i^{t_k}} \tag{2}$$

where $\hat{a}_i^s$ and $\hat{a}_i^{t_k}$ are the probability masses that belong to the probability simplex, i.e., $\sum_{i=1}^{n_s} \hat{a}_i^s = 1$ and $\sum_{i=1}^{n_{t_k}} \hat{a}_i^{t_k} = 1$.

Features generated from multitemporal SAR imagery can detect changes induced by human activities, natural disasters, etc. [40]. In particular, backscatter coefficient (σ°) and coherence coefficient (γ) are the most commonly used SAR features for change detection [41], [42]. The backscatter coefficient can show changes in surface roughness and target dielectric properties, while the coherence coefficient is very sensitive to small changes in the backscatterer's phase. Here, pre- and co-event σ° (i.e., $\sigma^\circ_{\text{pre}}$ and σ°_{co}) and pre- and co-event γ (i.e., γ_{pre} and γ_{co}) are used to extract the difference information for different land cover types in the changed areas. On this basis, we, respectively, define the BVF and CVF to filter the elements in D_{t_k} that are not relevant to the concerned geographic phenomena

$$\left\{x_i^{t_k} | \text{BVF}\left(\sigma^\circ_{\text{pre}}\left(x_i^{t_k}\right), \sigma^\circ_{\text{co}}\left(x_i^{t_k}\right)\right) \in \left(\alpha_1^k, \alpha_2^k\right)\right\} \tag{3}$$

$$\left\{x_i^{t_k} | \text{CVF}\left(\gamma_{\text{pre}}\left(x_i^{t_k}\right), \gamma_{\text{co}}\left(x_i^{t_k}\right)\right) \in \left(\beta_1^k, \beta_2^k\right)\right\}. \tag{4}$$

Specifically, there are pixel-based (such as difference, log-ratio, regression) [43] and texture-based (such as morphology operator, wavelets, co-occurrence matrix) [44] models available for setting BVF and CVF, which depend on specific

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change detection scenarios. The threshold parameters (i.e., α_1^k, α_2^k, β_1^k, and β_2^k) can be set based on ground reference values and empirical values from related studies.

After that, the updated data distribution of the subtarget domains ($\hat{\mu}_{t_k}$) can be obtained using (3) and/or (4). According to the application scenarios, it allows flexibility in determining whether to use σ° and γ in combination or only one of them.

Given the strong correlation between the elements of social media data (i.e., x_i^s) and the elements of prior data processed by land cover data and SAR features ($x_j^{t_k}$), the LCIOT adopts the geographical local correlation mode consistent with GOT [21], which is given as follows:

$$T\left(x_i^s\right) = \underset{x_j^{t_k} \in \hat{\mu}_{t_k}}{\arg\min}\, c\left(x_i^s, x_j^{t_k}\right) \tag{5}$$

where $c(x_i^s, x_j^{t_k})$ denotes the cost for transporting the probability mass from x_i^s to $x_j^{t_k}$ in the k subtarget domain

$$c\left(x_i^s, x_j^{t_k}\right) = \|x_i^s - x_j^{t_k}\|_2^2. \tag{6}$$

The calculation of transport cost in LCIOT only needs to consider the squared Euclidean distance between x_i^s and $x_j^{t_k}$, which is one of the main advantages over OT and GOT. There is no need to add any manual adjustment parameters between different feature costs. Note that since there are K subtarget domains, the source element x_i^s will be transported K times.

It can be observed that (5) is fully controlled by the cost $c(x_i^s, x_j^{t_k})$, leading to the possibility that multiple source elements may be transported to the same location. Here, we define $O(x_j^{t_k})$ as the total number of source elements transported to the same location in the subtarget domain. Let β be the overflow threshold. If $O(x_j^{t_k}) \leq \theta$, x_i^s is transported to $x_j^{t_k}$. Otherwise, if $O(x_j^{t_k}) > \theta$, the overflowing source elements will be transported to the next minimum cost location. In this study, θ is set to 1.

According to Tobler's first law of geography [45], areas closer to geotagged social media data are more likely to be flooded [46]. To identify the potential noise data far away from flood areas, the results of (5) are postprocessed as follows: If the transport distance from x_i^s to $x_j^{t_k}$ is larger than the given distance dist, this source element x_i^s is recognized to be noise data and removed. Here, dist is set to 1.5 km.

B. Application of LCIOT to Supervised Classification

The proposed LCIOT model can be easily applied to supervised classification tasks. Taking urban flood mapping as an example, we first extract intensity and coherence from SAR imagery to represent σ° and γ, respectively. Pixel-based subtraction operator is adopted as BVF and CVF considered in (3) and (4). The difference between pre- and co-event σ° and the difference between pre- and co-event γ are given as follows:

$$\text{DIFF-}\sigma^\circ = \sigma^\circ{}_{\text{co}} - \sigma^\circ{}_{\text{pre}} \tag{7}$$

$$\text{DIFF-}\gamma = \gamma_{\text{co}} - \gamma_{\text{pre}}. \tag{8}$$

Different land cover types exhibit distinct pre- and co-event SAR backscattering characteristics in urban flood events [47]. Specifically, the main backscattering mechanism is dihedral corner scattering, resulting in an increase in σ° [48]. Man-made structures (e.g., buildings) have high coherence when not flooded, while the floods between buildings can alter the spatial distribution of scatterers, causing a reduction in γ [47]. However, low-, medium-, and high-intensity areas in urban areas are characterized by different characteristics. The low-intensity areas exhibit a lower proportion of impervious surface and a higher proportion of vegetation, which generally leads to lower coherence, and tree canopies block dihedral scattering [49]. Therefore, the flooded low-intensity areas usually show a relatively small increase in σ° and a relatively small decrease in γ. The high-intensity areas have a higher proportion of impervious surface and a lower proportion of vegetation, which leads to a relatively large increase in σ° and a relatively large decrease in γ when flooded. When the high-intensity areas are smooth roads (e.g., highways), σ° maintains a low backscattering value and does not change significantly when flooded [47]. In this context, corresponding to (3) and (4), the threshold criteria of DIFF-σ° and DIFF-γ for low-, medium-, and high-intensity areas can be set as follows:

$$\left\{x_i^{t_l} | \text{DIFF-}\sigma^\circ\left(x_i^{t_l}\right) > \alpha^l \;\&\; \text{DIFF-}\gamma\left(x_i^{t_l}\right) < \beta^l\right\} \tag{9}$$

$$\left\{x_i^{t_m} | \text{DIFF-}\sigma^\circ\left(x_i^{t_m}\right) > \alpha^m \;\&\; \text{DIFF-}\gamma\left(x_i^{t_m}\right) < \beta^m\right\} \tag{10}$$

$$\left\{x_i^{t_h} | \text{DIFF-}\gamma\left(x_i^{t_h}\right) < \beta^h\right\} \tag{11}$$

where $x_i^{t_l}$, $x_i^{t_m}$, and $x_i^{t_h}$ are the elements in subtarget domain of low-, medium-, and high-intensity areas, respectively. The α^l and α^m are the threshold parameters of DIFF-σ° for low- and medium-intensity areas, respectively. The β^l, β^m, and β^h are the threshold parameters of DIFF-γ for low-, medium-, and high-intensity areas, respectively. According to the temporal variation analysis of σ° and γ from related studies on urban flood mapping [42], [49], here we roughly set the parameters of α^l, β^l, α^m, β^m, and β^h as 1, −0.1, 1.5, −0.15, and −0.2, respectively.

On this basis, the labeled flood category data for different land cover types can be obtained from (5). The labeled nonflood category data are provided through the original social media data due to the location uncertainty (most of which are located in nonflooded areas). To exclude a small part of the original data located in the flooded areas, DIFF-σ° and DIFF-γ are combined to filter the source elements by

$$\left\{x_i^s | \text{DIFF-}\sigma^\circ\left(x_i^s\right) < \alpha^s \;\&\; \text{DIFF-}\gamma\left(x_i^s\right) > \beta^s\right\}. \tag{12}$$

By referring to related studies [42], [49], the threshold parameters of α^s and β^s are roughly set as 2 and −0.1, respectively. Then, the supervised information of flood and nonflood categories can be used to train classification models for urban flood mapping.

Convolutional neural networks (CNNs) can automatically learn nonlinear patterns and features, which is widely used in urban flood mapping [50]. Here, we adopt the patch-based CNN as an example for the classification tasks of urban flood mapping, and other classifiers will provide a similar pattern. Compared with pixel-based CNN, patch-based CNN can reduce the impact of errors in multisequential image registration [51]. Specifically, the pre- and co-event SAR

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imagery is divided into nonoverlapping image patches of size 7 × 7.

In this study, a simple five-layer CNN model architecture is used, consisting of one input layer, three hidden layers, and one output layer. The hidden layers, respectively, contain 32, 64, and 128 neuron nodes using the ReLU activation function. The input data include pre-event intensity ($\sigma^{\circ}{}_{\text{pre}}$), co-event intensity ($\sigma^{\circ}{}_{\text{co}}$), pre-event coherence ($\gamma_{\text{pre}}$), and co-event coherence (γ_{co}). The image patches are labeled based on the aforementioned supervised information. Then, the labeled image patches are divided into a training dataset and a validation dataset according to a ratio of 9:1. We train the CNN model with a batch size of 64 and use the Adam optimizer with an initial learning rate of $1e^{-4}$. The number of epochs for the training of the CNN model is set to 200, and the model weights are saved every ten epochs. The best model is selected for urban flood mapping.

IV. Datasets

In August 2017, Hurricane Harvey made landfall in Texas and Louisiana, United States. The massive and extreme rainfall between August 26 and September 1 caused catastrophic flood disasters. According to statistics, the flood affected 13 million people, including 106 confirmed deaths and 39 000 displaced people, destroying more than 200 000 houses and causing economic losses of approximately $125 billion. The powerful convection of Hurricane Harvey has continued near the urban areas of Houston, Texas. Daily rainfall observed on August 26 and 27 reached 370 and 408 mm, respectively, which resulted in widespread flooding in Houston. In our experiment, three study areas located in western Houston are considered (Figs. S1–S3). Study area I and study area II contain social media data. The labeled data obtained through LCIOT from these two study areas is combined to provide supervised information for CNN model training. Finally, urban flood mapping and accuracy assessment are conducted for all study areas.

A. Remote Sensing Imagery

Concerning the Houston flood event in 2017, pre- and co-event Sentinel-1 SAR imagery is publicly available from the Copernicus Open Access Hub [52]. We collect Sentinel-1 ground range detected (GRD) products and single-look complex (SLC) products of the interferometric wide (IW) swath mode to calculate SAR intensity and SAR coherence, respectively (Table S1). The preprocessing steps for intensity data include the following: 1) orbit correction through accurate orbit files; 2) eliminating the thermal noise inherent in the SAR satellite system; 3) converting backscattered signals into backscattered signals through radiation calibration; 4) refined Lee speckle filter (window size of 7 × 7 pixels) for speckle reduction; and 5) range Doppler terrain correction for radiometric correction of terrain. The preprocessing steps for coherence data include the following: 1) orbit correction; 2) back geocoding of pre- and co-event images; 3) coherence estimation with a window size of 28 × 7 (range × azimuth); 4) deburst operation for removing black separation lines between bursts; 5) removing the terrain phase through the digital elevation model (DEM); 6) multilooking with a 4 × 1 window to get a square pixel; and 7) resampling the coherence data to 10 m.

B. Social Media Data

X, commonly referred to by its former name "Twitter," is a microblogging service platform that enables users to send and receive short messages in posts (formerly "tweets") [53]. Note that this study still uses the terms "Twitter" and "tweets" to be consistent with the existing research. Geotagging is the process of appending geographic coordinates to social media based on the global positioning system (GPS) of a mobile device, consisting of latitude and longitude information. In this study, geotagged Twitter data related to the 2017 Houston flood event are provided by [54]. Specifically, the geotagged Twitter data were gathered using keywords associated with Hurricane Harvey through the official Twitter Streaming application programming interface (API) and were spatiotemporally filtered across the affected areas of Hurricane Harvey. After that, a manual procedure for validation of geotagged Twitter data was performed. There are 112 and 47 geotagged tweets during the flood event in study area I [Fig. S1(c)] and study area II [Fig. S2(c)], respectively, while there are no geotagged tweets in the study area III.

C. Land Cover Data

The National Land Cover Database (NLCD) provides land cover data for the United States at 30-m spatial resolution through time-series Landsat imagery, which is freely available through the United States Geological Survey (USGS) [55]. The secondary classification system of the NLCD includes developed land types with varying proportions of impervious surface. Here, we use the land cover data of low-, medium-, and high-intensity developed areas for the study areas [Figs. S1(a), S2(a), and S3(a)].

D. Prior Flood Data and Ground Truth

The special flood hazard area (SFHA) shown on the flood insurance rate maps (FIRMs) provides an effective means to collect prior flood data in the study areas, which is defined as the area that will be inundated by the flood event (having a 1% chance of being equaled or exceeded in any given year) [56]. The SFHA for Houston is publicly available from the Federal Emergency Management Agency. Based on the very high-resolution optical imagery acquired by the National Oceanic and Atmospheric Administration (NOAA) on August 30 and 31, 2017, we visually interpret the flooded and non-flooded areas as ground truth to evaluate the experimental results [Figs. S1(b), S2(b), and S3(b)].

V. Results

In this section, we first quantitatively evaluate the precision of the relocated tweets obtained by different data fusion models for the 2017 Houston flood event. Then, accuracy assessment of urban flood mapping based on the supervised information generated by different models is performed.

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TABLE I
EVALUATION OF PRECISION AND CORRECTED PRECISION OF RELOCATED TWEETS OBTAINED BY DIFFERENT DATA FUSION MODELS

Models	TP	FP	Precision	Corrected precision (2 km)	Corrected precision (1.5 km)	Corrected precision (1 km)
None	28	131	17.61%	-	-	-
OT	126	33	79.25%	55.35%	44.03%	32.08%
GOT	156	3	98.11%	93.08%	78.62%	56.60%
LCIOT	322	3	99.08%	99.08%	99.08%	76.00%

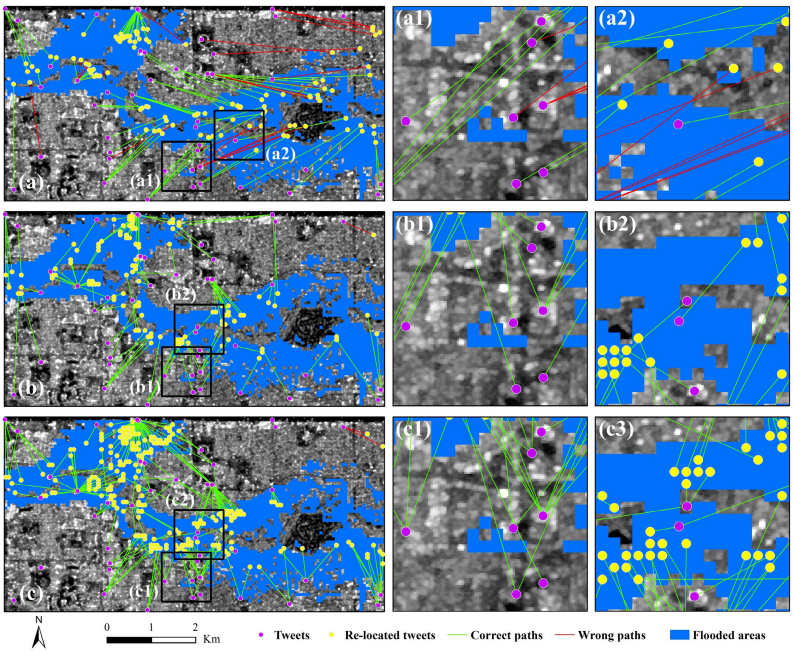


Fig. 2. Spatial distribution of relocated tweets obtained by (a) OT, (b) GOT, and (c) LCIOT.

A. Evaluation of Relocated Tweets

The metrics used for evaluating the relocated tweets are as follows.

1) "TP" denotes the number of relocated tweets that fall within the ground-truth flooded areas.
2) "FP" denotes the number of relocated tweets that fall outside the ground-truth flooded areas.
3) "Precision" denotes the proportion of TPs, which is given by Precision $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$.
4) "Corrected precision" denotes the proportion of relocated tweets that fall within the ground-truth flooded areas and the transport distance is less than a certain threshold.

Given that the probability that nearby social media data are related to flooding is much higher than that of distant social media data [46], three different distances are set up to calculate the corrected precision, which are 1, 1.5, and 2 km, respectively.

Table I presents the precision and corrected precision of relocated tweets obtained by different models. First, it can be found that the precision of original geotagged tweets is only 18%, which indicates most geotagged tweets are outside the flooded areas and have large location uncertainty. After the relocation of geotagged tweets by heterogeneous data fusion models, the location uncertainty is alleviated. Specifically, LCIOT and GOT have very high precision in relocated tweets, with, respectively, 99% and 98% of the geotagged tweets aligned to the flooded areas, while the precision of OT is relatively low (79%). Regarding the corrected precision considering different distances, LCIOT outperforms OT and GOT. The corrected precision (2 km) of OT, GOT, and LCIOT is 99%, 93%, and 55%, respectively. The corrected precision (1.5 km) of OT, GOT, and LCIOT is 99%, 79%, and 44%, respectively. The corrected precision (1 km) of OT, GOT, and LCIOT is 76%, 57%, and 32%, respectively. Overall, the proposed LCIOT can better align the geotagged tweets to the flooded areas near Twitter users.

In addition, the spatial distribution of relocated tweets and the transport paths obtained by different models are intuitively shown in Figs. 2 and S4. It can be observed that the proposed LCIOT obtains more relocated tweets (325) than OT (159) and GOT (159). This is because the LCIOT is able to align geotagged tweets to the flooded areas for different land cover types. Meanwhile, LCIOT can identify the potential noise data far away from flood areas, thereby some geotagged tweets are not transported to the SFHA. Furthermore, Fig. 3 shows the distribution of transport distance and average transport

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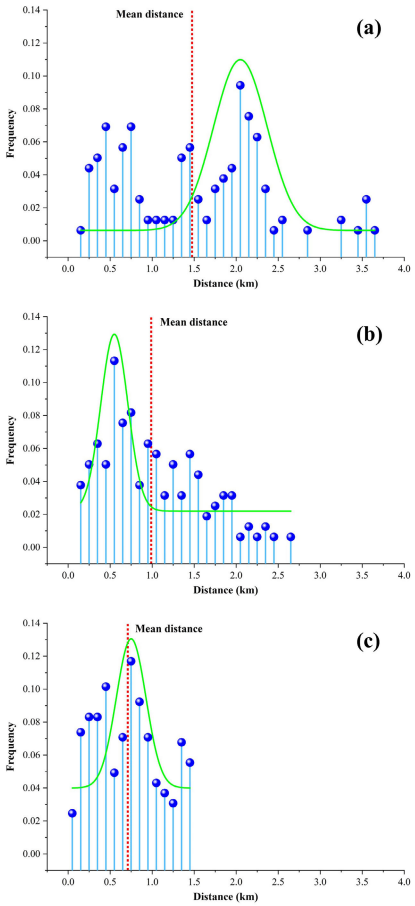


Fig. 3. Distribution of transport distance obtained by (a) OT, (b) GOT, and (c) LCIOT.

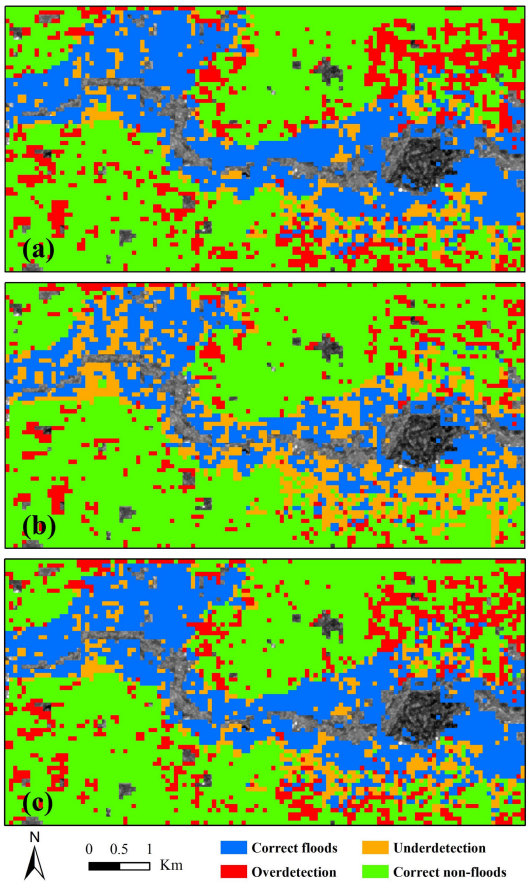


Fig. 4. Urban flood mapping combining Sentinel-1 SAR imagery and supervised information obtained by (a) OT, (b) GOT, and (c) LCIOT.

distance obtained by different models. The average transport distances of LCIOT, GOT, and OT are 0.71, 0.99, and 1.48 km, respectively. This again shows that LCIOT is more effective in aligning the geotagged tweets from nonflooded areas to nearby flooded areas.

Based on the evaluation of relocated tweets obtained by different models, it can be concluded that compared with traditional OT and GOT models, the proposed LCIOT model is able to align the geotagged tweets to the nearby flooded areas with robust performance. And the relocated tweets obtained by LCIOT can provide labeled flood category data for different land cover types, i.e., low-, medium-, and high-intensity developed areas in this study.

B. Accuracy Assessment of Urban Flood Mapping

Based on the supervised information obtained by different data fusion models, such as OT, GOT, and LCIOT, a patch-based CNN classification model is adopted for urban flood mapping. To assess the classification results, the following quantitative metrics are calculated by averaging ten independent Monte Carlo runs, including precision, recall, $F1$ score, overall accuracy (OA), and Cohen's Kappa coefficient. With reference to relevant studies on urban flood mapping [47], [57], [58], a significance level of 0.05 is selected to estimate the confidence intervals of the metrics.

Table II presents the classification results with 95% confidence intervals. First, the OA and Kappa of LCIOT in the study areas range from 0.77 to 0.82 and 0.51 to 0.61, respectively, which are higher than those obtained by OT and GOT. An interesting finding is also derived from the results of OT and GOT, that is, OT has obviously low precision but high recall, while GOT has obviously high precision but low recall. This phenomenon can be explored through the data fusion mechanisms of the OT and GOT models. For OT, it is assumed that the probability mass of the source domain and the target domain follows a uniform distribution, leading to some source elements being relocated to locations outside the target domain. However, these locations are likely to be nonflooded areas. Therefore, the samples of the flood category are mixed with nonflood data, resulting in an overestimation of the flood extent [Figs. 4(a), S5(a), S6(a)]. For GOT, it tends to relocate the source elements to extreme flooded locations in the target domain, so there is a lack of labeled data for low- and medium-level flooding in the flood category, resulting in an underestimation of the flood extent [Figs. 4(b), S5(b), and S6(b)]. In contrast, the classification results of LCIOT are more accurate and reasonable [Figs. 4(c), S5(c), and S6(c)].

VI. Discussion

A. Reliability of the Heterogeneous Data Fusion Model

In this study, we put forward a general data fusion model for heterogeneous SAR imagery and social media data, namely,

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TABLE II
ACCURACY ASSESSMENT OF URBAN FLOOD MAPPING WITH A 95% CONFIDENCE INTERVAL USING CNNS AND SUPERVISED INFORMATION OBTAINED BY DIFFERENT DATA FUSION MODELS

Study area	Models	Precision	Recall	F1	OA	Kappa
Study area I	OT	0.649±0.013	0.739±0.021	0.69±0.004	0.761±0.005	0.498±0.007
	GOT	0.685±0.016	0.543±0.028	0.604±0.012	0.744±0.003	0.42±0.009
	LCIOT	0.661±0.01	0.738±0.014	0.697±0.003	0.769±0.004	0.512±0.006
	GOT	0.838±0.021	0.601±0.022	0.699±0.01	0.801±0.005	0.536±0.11
	LCIOT	0.759±0.016	0.766±0.015	0.762±0.003	0.815±0.005	0.611±0.008
Study area III	OT	0.638±0.022	0.845±0.02	0.725±0.008	0.738±0.016	0.484±0.026
	GOT	0.842±0.031	0.516±0.045	0.635±0.025	0.76±0.007	0.471±0.021
	LCIOT	0.748±0.023	0.739±0.026	0.742±0.004	0.789±0.007	0.564±0.01

TABLE III
ACCURACY ASSESSMENT OF URBAN FLOOD MAPPING WITH A 95% CONFIDENCE INTERVAL USING SVMS AND SUPERVISED INFORMATION OBTAINED BY DIFFERENT DATA FUSION MODELS

Study area	Models	Precision	Recall	F1	OA	Kappa
Study area I	OT	0.643±0.003	0.824±0.007	0.722±0.002	0.744±0.002	0.492±0.003
	GOT	0.691±0.007	0.656±0.018	0.673±0.009	0.743±0.004	0.462±0.01
	LCIOT	0.677±0.003	0.807±0.005	0.736±0.001	0.767±0.001	0.53±0.002
Study area III	OT	0.646±0.006	0.886±0.005	0.747±0.005	0.73±0.006	0.472±0.011
	GOT	0.789±0.002	0.608±0.006	0.686±0.004	0.749±0.002	0.484±0.005
	LCIOT	0.695±0.004	0.837±0.009	0.759±0.002	0.761±0.002	0.526±0.003
Study area III	OT	0.677±0.008	0.909±0.021	0.774±0.006	0.725±0.005	0.442±0.011
	GOT	0.786±0.011	0.661±0.02	0.715±0.008	0.728±0.003	0.459±0.005
	LCIOT	0.717±0.008	0.833±0.018	0.769±0.004	0.74±0.001	0.476±0.002

LCIOT. Different from the traditional OT [11] and GOT [21] designed for the fusion of optical imagery and social media data, the proposed LCIOT is the first attempt to fuse SAR imagery and social media data. The main advantages of LCIOT over OT and GOT are summarized as follows: 1) the LCIOT considers the one-to-many data association based on the variations in backscatter and coherence for different land cover types; 2) the relocation of geotagged tweets is improved by excluding noise data unrelated to concerned geographic phenomena; and 3) the operating efficiency for heterogeneous data fusion can be improved due to the reduction in the number of elements in the target domain. The CNN results of urban flood mapping also prove the effectiveness of the proposed LCIOT (Table II, Fig. 4). On this basis, we further test the supervised information obtained by different data fusion models for urban flood mapping using a standard classifier, such as support vector machines (SVMs) [59]. Table III presents the SVM results of urban flood mapping with a 95% confidence interval. The proposed LCIOT outperforms OT and GOT in all study areas. It can also be discovered that OT has low precision but high recall, whereas GOT has high precision but low recall, which is consistent with the findings in CNN results. This implies that the results of urban flood mapping based on OT and GOT are easily affected by the area ratio between flood areas and nonflood areas within a spatial extent. In addition, we compare the accuracy results of urban flood maps produced with CNN and SVM (Table S2). SVM have better results in study area I and $F1$ score, while CNN performs better in study area II and study area III as well as OA and Kappa coefficient. Overall, the supervised information obtained by LCIOT is effective for urban flood mapping using deep learning models (e.g., CNN) or machine learning models (e.g., SVM).

B. Generalization of the Proposed Model

The proposed LCIOT can be generalized to a variety of scenarios or applications. For instance, besides Sentinel-1 imagery, the SAR data from other sensors can be used as the input for the proposed model, such as TerraSAR-X [60] and RADARSAT-2 [61]. Social media data from other social networks can also be considered, such as Facebook, Instagram, YouTube, and Weibo, among others [62]. Concerning the prior flood data, many countries and regions have generated flood risk maps based on information on historical flood events, hydrological station data, and hydrological models, such as the United States [56], the European Union [63], and Japan [64]. Meanwhile, several flood risk datasets have been developed with global coverage, which can provide prior information on flooding at the national scale [65]. As for land cover data, many global land cover datasets are publicly available and can be used as model inputs [66]. In addition, the generalization of the model needs to consider parameter settings. Taking the SAR backscattering characteristics

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TABLE IV
EVALUATION OF THE PARAMETER SETTINGS IN THE LCIOT

Scenarios	TP	FP	Precision	Corrected precision (2 km)	Corrected precision (1.5 km)	Corrected precision (1 km)
Scenario I	312	3	99.05%	99.05%	99.05%	75.24%
Scenario II	322	3	99.08%	99.08%	99.08%	76.00%
Scenario III	363	5	98.64%	98.64%	98.64%	77.72%

Scenario I: α^l = 0.5, β^l = -0.05, α^m = 1, β^m = -0.1, β^h = -0.15. Scenario II: α^l = 1, β^l = -0.1, α^m = 1.5, β^m = -0.15, β^h = -0.2. Scenario III: α^l = 1.5, β^l = -0.15, α^m = 2, β^m = -0.2, β^h = -0.25.

TABLE V
IMPACT OF TIME DIFFERENCE BETWEEN SAR IMAGERY AND SOCIAL MEDIA DATA ON DATA FUSION

Date	Aug 26	Aug 27	Aug 28	Aug 29	Aug 30	Aug 31	Sep 1
Corrected precision (1 km)	60.00%	62.96%	71.88%	77.78%	82.00%	80.85%	76.90%
Transport distance (m)	823.75	819.89	805.46	720.92	595.46	712.48	773.60

of different land cover types as an example, the parameter settings of (9)–(11) should consider factors such as sensors, temporal baselines, and spatial baseline [47]. In general, the parameters can be set according to the feature analysis of similar scenarios or applications in related studies [42], [49] or statistical analysis of ground reference data [67]. Furthermore, we evaluate the LCIOT under different parameter settings of DIFF-σ° and DIFF-γ (Table IV). Scenario II corresponds to the parameter settings adopted in this study, while Scenarios I and III correspond to parameter settings under more relaxed and stricter criteria for flood feature extraction, respectively. It is discovered that the LCIOT consistently maintains robust performance.

C. Potential Sources of Uncertainty and Future Work

This study mainly focuses on the issues of discrepancies in data distribution and geolocation uncertainty. However, since remote sensing generates observation data at a certain time and social media obtain continuous observation data, there might be difference in information regarding the same geographical phenomenon [68], which could affect data fusion and model application. In view of the potential impact of time difference, we evaluate the relocation accuracy and transport distance of geotagged Twitter data collected during the 2017 Houston flood event in Study Area I (Table V). Note that the time range of geotagged Twitter data is from August 26 to September 1, 2017, while the acquisition time of pre-event Sentinel-1 SAR imagery is August 30, 2017. The evaluation findings indicate that the smaller the time difference between the data, the higher the corrected precision and the shorter the transport distance. For instance, on August 30 and August 26, the corrected precision is 82% and 60%, respectively, and the transport distances are 595 and 824 m, respectively. This may be related to the changes in the runoff characteristics of urban floods and the mixed drainage system of natural and artificial drainage [69], [70]. Although the temporal difference between Sentinel-1 SAR imagery and geotagged Twitter data is small in the 2017 Houston flood event, we also take some postprocessing to lessen the possibility of erroneous information association caused by the temporal difference, such as excluding data with a transport distance exceeding a certain range. In general, when the temporal difference is small or for geographical phenomena that do not change significantly over time (e.g., socioeconomic factor estimation [71], urban land use classification [72]), the impact of temporal difference can be neglected to a certain extent. When there is a significant temporal difference or for geographical phenomena that change dynamically over time, it is more necessary to consider the impact of temporal difference on data fusion. In future research, the temporal correlation of heterogeneous data can be taken into account in the proposed LCIOT model through techniques such as dynamic time warping (DTW) [73]. In another aspect, multiple sources of geotagged social media data should be considered. For instance, location information can be extracted from social media text, photographs, and videos through NLP and visual place recognition to enrich geotagged social media data [9], [74]. In addition, volunteered geographic information (VGI) and in situ sensor systems can be used as supplements [75].

VII. CONCLUSION

In this article, a general data fusion model called LCIOT is developed for the fusion of heterogeneous SAR imagery and social media data. Taking the 2017 Houston flood event as a case study, we evaluate the proposed LCIOT model and its application to supervised classification tasks. According to our experiments, the LCIOT can align the geotagged tweets from nonflooded areas to nearby flooded areas with high precision, outperforming the traditional OT and GOT models. Based on the fusion results of LCIOT, supervised information for different land cover types when flooded can be obtained for supervised classifiers such as CNN and SVM. Accurate and reasonable results for urban flood mapping are realized by the proposed LCIOT. In conclusion, the LCIOT provides a general framework for heterogeneous data fusion, which has potential for a variety of scenarios and applications.

REFERENCES

[1] P. Gamba, "Human settlements: A global challenge for EO data processing and interpretation," *Proc. IEEE*, vol. 101, no. 3, pp. 570–581, Mar. 2013.

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[2] J. Li et al., "Social media: New perspectives to improve remote sensing for emergency response," *Proc. IEEE*, vol. 105, no. 10, pp. 1900–1912, Oct. 2017.

[3] V. Kotovirta, T. Toivanen, R. Tergujeff, T. Häme, and M. Molinier, "Citizen science for earth observation: Applications in environmental monitoring and disaster response," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 40, pp. 1221–1226, Apr. 2015.

[4] G. Cheng, J. Han, L. Guo, Z. Liu, S. Bu, and J. Ren, "Effective and efficient midlevel visual elements-oriented land-use classification using VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4238–4249, Aug. 2015.

[5] S. S. Oliveira, B. Barros, J. L. Pereira, P. T. Santos, and R. Pereira, "Social media use by citizen science projects: Characterization and recommendations," *Frontiers Environ. Sci.*, vol. 9, Jul. 2021, Art. no. 715319.

[6] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[7] Z. Liu, J. Li, L. Wang, and A. Plaza, "Integration of remote sensing and crowdsourced data for fine-grained urban flood detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13523–13532, 2024.

[8] D. C. Mason, M. S. Horritt, J. T. Dall'Amico, T. R. Scott, and P. D. Bates, "Improving river flood extent delineation from synthetic aperture radar using airborne laser altimetry," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 3932–3943, Dec. 2007.

[9] J. Li, Z. Liu, X. Lei, and L. Wang, "Distributed fusion of heterogeneous remote sensing and social media data: A review and new developments," *Proc. IEEE*, vol. 109, no. 8, pp. 1350–1363, Aug. 2021.

[10] Y. George, S. Karunasekera, A. Harwood, and K. H. Lim, "Real-time spatio-temporal event detection on geotagged social media," *J. Big Data*, vol. 8, no. 1, p. 91, Dec. 2021.

[11] H. Wang, E. Skau, H. Krim, and G. Cervone, "Fusing heterogeneous data: A case for remote sensing and social media," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 6956–6968, Dec. 2018.

[12] Z. Liu, J. Li, J. Plaza, and A. Plaza, "Subspace optimal transport for spatial bias correction of social media data: A case study of 2013 Boulder flood event," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 3249–3252.

[13] N. Courty et al., "Optimal transport for domain adaptation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 9, pp. 1853–1865, Sep. 2017.

[14] L. Zhang, J. He, Q. Yang, Y. Xiao, and Q. Yuan, "Data-driven multi-source remote sensing data fusion: Progress and challenges," *Acta Geodaetica et Cartographica Sinica*, vol. 51, no. 7, p. 1317, 2022.

[15] A. Kusumo, D. Reckien, and J. Verplanke, "Utilising volunteered geographic information to assess resident's flood evacuation shelters. Case study: Jakarta," *Appl. Geography*, vol. 88, pp. 174–185, Nov. 2017.

[16] L. Smith, Q. Liang, P. James, and W. Lin, "Assessing the utility of social media as a data source for flood risk management using a real-time modelling framework," *J. Flood Risk Manage.*, vol. 10, no. 3, pp. 370–380, 2017.

[17] C. Songchon, G. Wright, and L. Beevers, "Quality assessment of crowdsourced social media data for urban flood management," *Comput., Environ. Urban Syst.*, vol. 90, Nov. 2021, Art. no. 101690.

[18] Z. Liu, J. Li, J. Plaza, and A. Plaza, "Spatial bias correction of social media data by exploiting remote sensing knowledge in data-deficient regions," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 7017–7020.

[19] I. Redko, E. Morvant, A. Habrard, M. Sebban, and Y. Bennani, *Advances in Domain Adaptation Theory*. Amsterdam, The Netherlands: Elsevier, 2019.

[20] D. Tuia, C. Persello, and L. Bruzzone, "Domain adaptation for the classification of remote sensing data: An overview of recent advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 41–57, Jun. 2016.

[21] Z. Liu, Q. Qiu, J. Li, L. Wang, and A. Plaza, "Geographic optimal transport for heterogeneous data: Fusing remote sensing and social media," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6935–6945, Aug. 2021.

[22] Z. Lipton, Y.-X. Wang, and A. Smola, "Detecting and correcting for label shift with black box predictors," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 3122–3130.

[23] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2960–2967.

[24] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.

[25] L. Caicedo Torres, L. Manella Pereira, and M. Hadi Amini, "A survey on optimal transport for machine learning: Theory and applications," 2021, *arXiv:2106.01963*.

[26] V. Titouan, N. Courty, R. Tavenard, and R. Flamary, "Optimal transport for structured data with application on graphs," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6275–6284.

[27] J. Wu, W. Han, J. Chen, and S. Wang, "Improving geological remote sensing interpretation via optimal transport-based point–surface data fusion," *Remote Sens.*, vol. 16, no. 1, p. 53, Dec. 2023.

[28] J. Mifdal, B. Coll, N. Courty, J. Froment, and B. Vedel, "Hyperspectral and multispectral Wasserstein barycenter for image fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3373–3376.

[29] M. N. I. Sarker, Y. Peng, C. Yiran, and R. C. Shouse, "Disaster resilience through big data: Way to environmental sustainability," *Int. J. Disaster Risk Reduction*, vol. 51, Dec. 2020, Art. no. 101769.

[30] N. Zhao, W. Zhang, Y. Liu, E. L. Samson, Y. Chen, and G. Cao, "Improving nighttime light imagery with location-based social media data," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2161–2172, Apr. 2019.

[31] S. Du, S. Du, B. Liu, X. Zhang, and Z. Zheng, "Large-scale urban functional zone mapping by integrating remote sensing images and open social data," *GIScience Remote Sens.*, vol. 57, no. 3, pp. 411–430, 2020.

[32] B. Bischke, D. Borth, C. Schulze, and A. Dengel, "Contextual enrichment of remote-sensed events with social media streams," in *Proc. 24th ACM Int. Conf. Multimedia*, Oct. 2016, pp. 1077–1081.

[33] L. Xu and A. Ma, "Coarse-to-fine waterlogging probability assessment based on remote sensing image and social media data," *Geo-spatial Inf. Sci.*, vol. 24, no. 2, pp. 279–301, Apr. 2021.

[34] K. Ahmad, K. Pogorelov, M. Riegler, N. Conci, and P. Halvorsen, "Social media and satellites: Disaster event detection, linking and summarization," *Multimedia Tools Appl.*, vol. 78, no. 3, pp. 2837–2875, Feb. 2019.

[35] Q. Huang, G. Cervone, and G. Zhang, "A cloud-enabled automatic disaster analysis system of multi-sourced data streams: An example synthesizing social media, remote sensing and Wikipedia data," *Comput., Environ. Urban Syst.*, vol. 66, pp. 23–37, Nov. 2017.

[36] E. Schnebele, G. Cervone, S. Kumar, and N. Waters, "Real time estimation of the calgary floods using limited remote sensing data," *Water*, vol. 6, no. 2, pp. 381–398, Feb. 2014.

[37] R. Sadiq, Z. Akhtar, M. Imran, and F. Ofli, "Integrating remote sensing and social sensing for flood mapping," *Remote Sens. Appl., Soc. Environ.*, vol. 25, Jan. 2022, Art. no. 100697.

[38] E. Schnebele and G. Cervone, "Improving remote sensing flood assessment using volunteered geographical data," *Natural Hazards Earth Syst. Sci.*, vol. 13, no. 3, pp. 669–677, Mar. 2013.

[39] G. Cervone, E. Schnebele, N. Waters, M. Moccaldi, and R. Sicignano, "Using social media and satellite data for damage assessment in urban areas during emergencies," in *Seeing Cities Through Big Data: Research, Methods and Applications in Urban Informatics*, 2017, pp. 443–457.

[40] C. Marin, F. Bovolo, and L. Bruzzone, "Building change detection in multitemporal very high resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2664–2682, May 2015.

[41] Q. Zhao et al., "Understanding the temporal dynamics of coherence and backscattering using Sentinel-1 imagery for crop-type mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6875–6893, 2024.

[42] M. Chini, R. Pelich, L. Pulvirenti, N. Pierdicca, R. Hostache, and P. Matgen, "Sentinel-1 InSAR coherence to detect floodwater in urban areas: Houston and hurricane Harvey as a test case," *Remote Sens.*, vol. 11, no. 2, p. 107, Jan. 2019.

[43] R. Zhao, G. H. Peng, W. Yan, L. L. Pan, and L. Y. Wang, "Change detection in SAR images based on superpixel segmentation and image regression," *Earth Sci. Inform.*, vol. 14, pp. 69–79, Nov. 2021.

[44] O. Ajadi, F. Meyer, and P. Webley, "Change detection in synthetic aperture radar images using a multiscale-driven approach," *Remote Sens.*, vol. 8, no. 6, p. 482, Jun. 2016.

[45] W. R. Tobler, "A computer movie simulating urban growth in the Detroit region," *Econ. Geogr.*, vol. 46, pp. 234–240, Jun. 1970.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:50:38 UTC from IEEE Xplore. Restrictions apply.

[46] J. P. De Albuquerque, B. Herfort, A. Brenning, and A. Zipf, "A geographic approach for combining social media and authoritative data towards identifying useful information for disaster management," *Int. J. Geographical Inf. Sci.*, vol. 29, no. 4, pp. 667–689, 2015.

[47] Y. Li, S. Martinis, and M. Wieland, "Urban flood mapping with an active self-learning convolutional neural network based on TerraSAR-X intensity and interferometric coherence," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 178–191, Jun. 2019.

[48] D. Amitrano et al., "Urban areas enhancement in multitemporal SAR RGB images using adaptive coherence window and texture information," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3740–3752, Aug. 2016.

[49] Y. Li, S. Martinis, M. Wieland, S. Schlaffer, and R. Natsuaki, "Urban flood mapping using SAR intensity and interferometric coherence via Bayesian network fusion," *Remote Sens.*, vol. 11, no. 19, p. 2231, Sep. 2019.

[50] A. Gebrehiwot, L. Hashemi-Beni, G. Thompson, P. Kordjamshidi, and T. E. Langan, "Deep convolutional neural network for flood extent mapping using unmanned aerial vehicles data," *Sensors*, vol. 19, no. 7, p. 1486, Mar. 2019.

[51] B. Peng, Z. Meng, Q. Huang, and C. Wang, "Patch similarity convolutional neural network for urban flood extent mapping using bi-temporal satellite multispectral imagery," *Remote Sens.*, vol. 11, no. 21, p. 2492, Oct. 2019.

[52] M. Fernández et al., "Copernicus Sentinel–1 POD reprocessing campaign," *Adv. Space Res.*, vol. 70, no. 2, pp. 249–267, 2022.

[53] D. Murthy, "Sociology of Twitter/X: Trends, challenges, and future research directions," *Annu. Rev. Sociol.*, vol. 50, no. 1, pp. 169–190, Aug. 2024.

[54] C. Guo and Q. Huang, "Examining spatiotemporal and sentiment patterns of evacuation behavior during 2017 hurricane Harvey," in *Empowering Human Dynamics Research with Social Media and Geospatial Data Analytics*. Cham, Switzerland: Springer, 2021, pp. 139–165.

[55] S. Jin et al., "Overall methodology design for the United States national land cover database 2016 products," *Remote Sens.*, vol. 11, no. 24, p. 2971, Dec. 2019.

[56] R. Blessing, A. Sebastian, and S. D. Brody, "Flood risk delineation in the United States: How much loss are we capturing?" *Natural Hazards Rev.*, vol. 18, no. 3, Aug. 2017, Art. no. 04017002.

[57] P. Olofsson, G. M. Foody, M. Herold, S. V. Stehman, C. E. Woodcock, and M. A. Wulder, "Good practices for estimating area and assessing accuracy of land change," *Remote Sens. Environ.*, vol. 148, pp. 42–57, May 2014.

[58] V. Gude, S. Corns, and S. Long, "Flood prediction and uncertainty estimation using deep learning," *Water*, vol. 12, no. 3, p. 884, Mar. 2020.

[59] D. A. Pisner and D. M. Schnyer, "Support vector machine," in *Machine Learning*. Amsterdam, The Netherlands: Elsevier, 2020, pp. 101–121.

[60] D. C. Mason, R. Speck, B. Devereux, G. J.-P. Schumann, J. C. Neal, and P. D. Bates, "Flood detection in urban areas using TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 2, pp. 882–894, Feb. 2010.

[61] M. Tanguy, K. Chokmani, M. Bernier, J. Poulin, and S. Raymond, "River flood mapping in urban areas combining Radarsat-2 data and flood return period data," *Remote Sens. Environ.*, vol. 198, pp. 442–459, Sep. 2017.

[62] J. Guo, Y. Du, and C. Fan, "Inferring cascading effects of natural and built environments failures from social media videos in disasters," *Int. J. Disaster Risk Reduction*, vol. 104, Apr. 2024, Art. no. 104378.

[63] H. De Moel, J. Van Alphen, and J. C. Aerts, "Flood maps in Europe—Methods, availability and use," *Natural Hazards Earth Syst. Sci.*, vol. 9, no. 2, pp. 289–301, 2009.

[64] I. Nakamura and M. C. Llasat, "Policy and systems of flood risk management: A comparative study between Japan and Spain," *Natural Hazards*, vol. 87, no. 2, pp. 919–943, Jun. 2017.

[65] M. V. Bernhofen et al., "The role of global data sets for riverine flood risk management at national scales," *Water Resour. Res.*, vol. 58, no. 4, 2022, Art. no. e2021WR031555.

[66] N. Tsendbazar, S. De Bruin, and M. Herold, "Assessing global land cover reference datasets for different user communities," *ISPRS J. Photogramm. Remote Sens.*, vol. 103, pp. 93–114, May 2015.

[67] P. Manjusree, L. Prasanna Kumar, C. M. Bhatt, G. S. Rao, and V. Bhanumurthy, "Optimization of threshold ranges for rapid flood inundation mapping by evaluating backscatter profiles of high incidence angle SAR images," *Int. J. Disaster Risk Sci.*, vol. 3, pp. 113–122, Aug. 2012.

[68] G. Panteras and G. Cervone, "Enhancing the temporal resolution of satellite-based flood extent generation using crowdsourced data for disaster monitoring," *Int. J. Remote Sens.*, vol. 39, no. 5, pp. 1459–1474, Mar. 2018.

[69] F. Piadeh, K. Behzadian, and A. M. Alani, "A critical review of real-time modelling of flood forecasting in urban drainage systems," *J. Hydrol.*, vol. 607, Apr. 2022, Art. no. 127476.

[70] H. Huang, X. Wang, and L. Liu, "A review on urban pluvial floods: Characteristics, mechanisms, data, and research methods," *Prog. Geography*, vol. 40, no. 6, pp. 1048–1059, 2021.

[71] N. Zhao, G. Cao, W. Zhang, and E. L. Samson, "Tweets or nighttime lights: Comparison for preeminence in estimating socioeconomic factors," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 1–10, Dec. 2018.

[72] L. Qi, J. Li, Y. Wang, and X. Gao, "Urban observation: Integration of remote sensing and social media data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 11, pp. 4252–4264, Nov. 2019.

[73] H. Janati, M. Cuturi, and A. Gramfort, "Spatio-temporal alignments: Optimal transport through space and time," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2020, pp. 1695–1704.

[74] G. Cervone, E. Sava, Q. Huang, E. Schnebele, J. Harrison, and N. Waters, "Using Twitter for tasking remote-sensing data collection and damage assessment: 2013 Boulder flood case study," *Int. J. Remote Sens.*, vol. 37, no. 1, pp. 100–124, Jan. 2016.

[75] H. R. Fazeli, M. N. Said, S. Amerudin, and M. Z. A. Rahman, "A study of volunteered geographic information (VGI) assessment methods for flood hazard mapping: A review," *Jurnal Teknologi*, vol. 75, no. 10, pp. 127–134, 2015.



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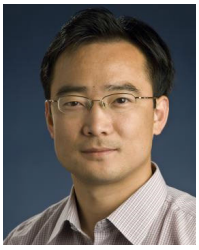


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